

# ELECENG 2E14: Electronic Devices and Circuits 1

## Design Project 5

Anushka Chauhan - Chauha30 - 400598874

As a future member of the engineering profession, the student is responsible for performing the required work in an honest manner, without plagiarism and cheating. Submitting this work with my name and student number is a statement and understanding that this work is my own and adheres to the Academic Integrity Policy of McMaster University and the Code of Conduct of the Professional Engineers of Ontario. Submitted by [Anushka Chauhan, Chauha30 ,400598874]

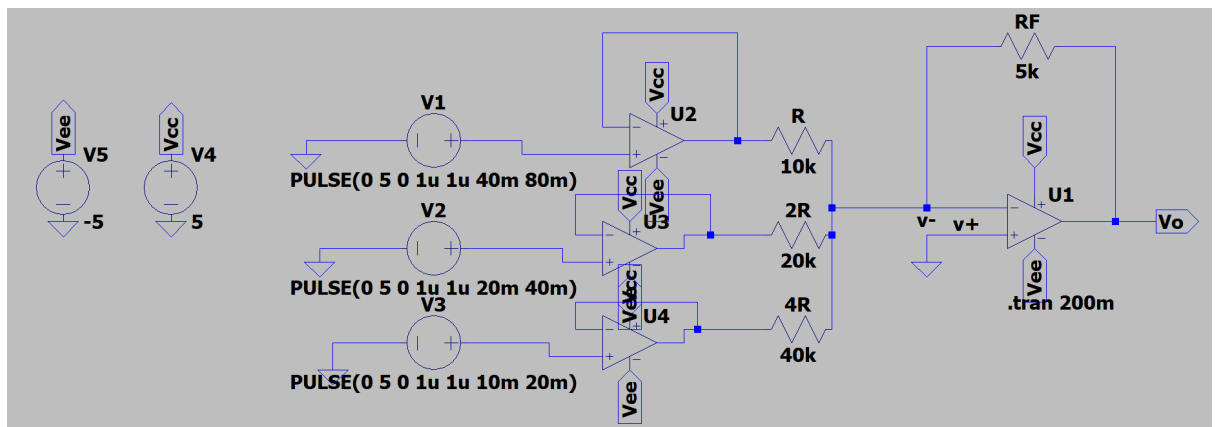
|                                                         |          |
|---------------------------------------------------------|----------|
| <b>ELECENG 2EI4: Electronic Devices and Circuits 1</b>  | <b>1</b> |
| <b>Design Project 5</b>                                 | <b>1</b> |
| Summary                                                 | 3        |
| Design                                                  | 3        |
| a. Circuit Diagram and Simulation Result                | 3        |
| b. Description of how the circuit works                 | 4        |
| c. Calculations                                         | 5        |
| d. Reference for the Circuit                            | 6        |
| Measurement and Analysis                                | 7        |
| a. Physical Circuit                                     | 7        |
| b. Description of how the digital inputs were generated | 8        |
| c. Transfer Characteristics of the Circuit              | 9        |
| d. Gain Error                                           | 9        |
| e. Maximum Differential Non-Linearity                   | 10       |
| f. Offset Error                                         | 10       |
| Discussion                                              | 11       |
| a. Origin of various errors measured                    | 11       |
| b. Design Choice and Alternate Design                   | 11       |
| References                                              | 11       |

## Summary

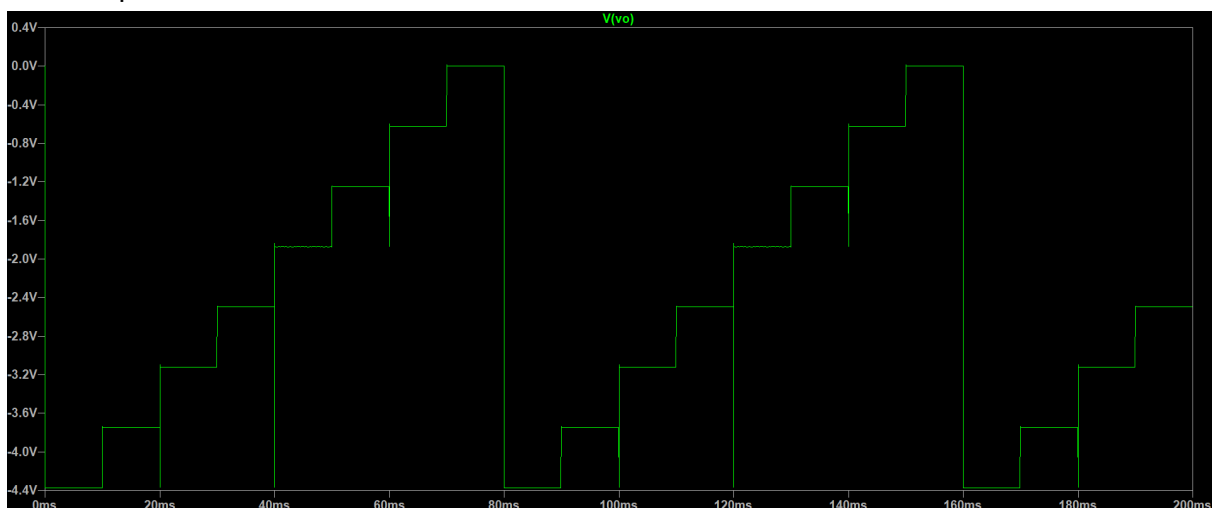
The objective of this project is to design and implement a 3-bit Digital-to-Analog Converter (DAC) with a full-scale analog voltage of 5V. The design must interface with digital logic levels of 0V (Low) and 5V (High) while strictly limiting the current drawn from the digital source to less than  $1\mu\text{A}$ . The design uses a special circuit called a Binary Weighted Resistor setup with an operational amplifier (Op-Amp). To keep the input signals from affecting the rest of the circuit, three buffer amplifiers are added. These buffers act like shields to separate the digital input from the rest of the circuit. The output of the circuit creates a staircase-like voltage that corresponds to a 3-bit binary input, meaning it can represent values from 0 to 7.

## Design

### a. Circuit Diagram and Simulation Result



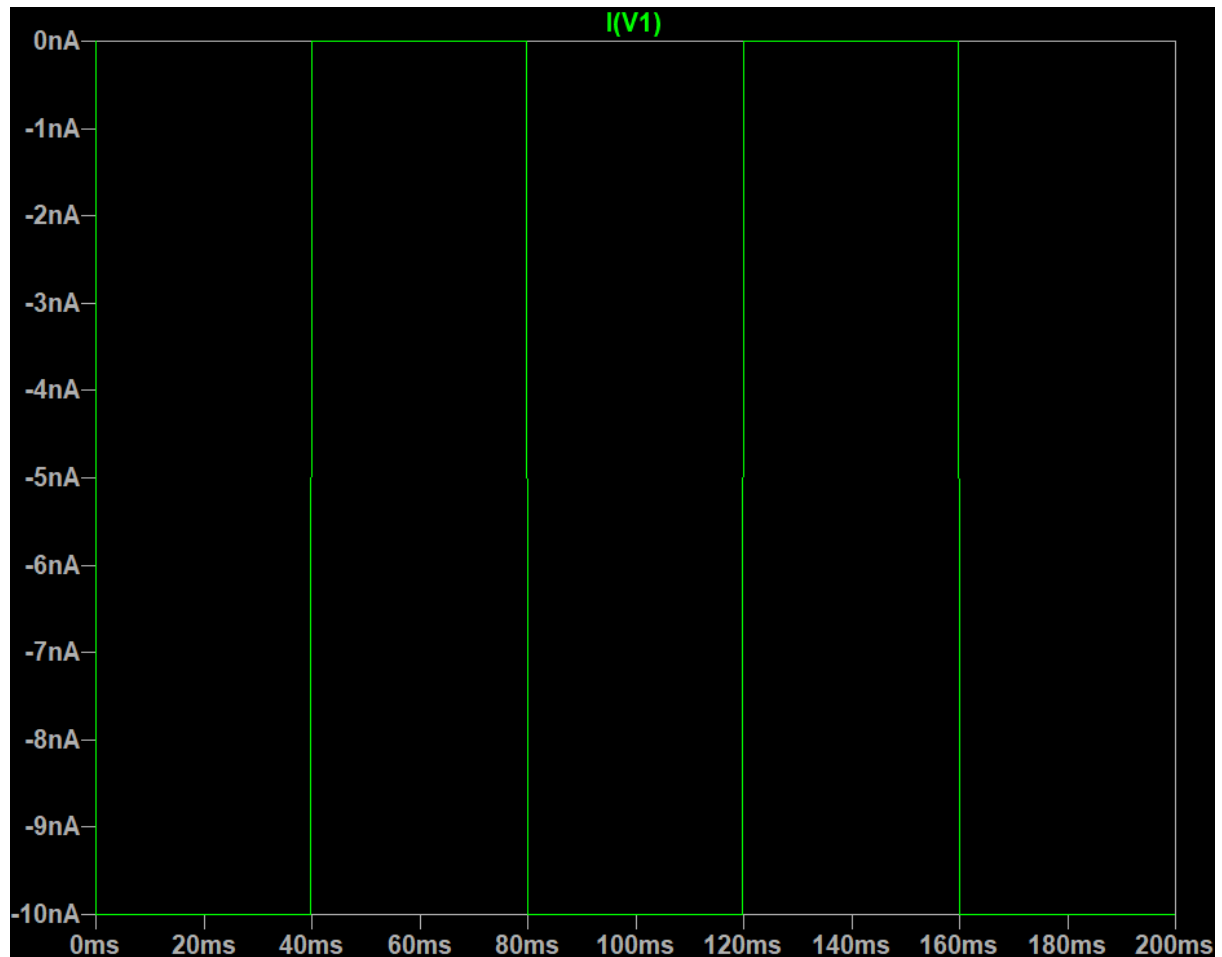
Vout Graph



While the LTspice and AD3 waveforms look different, they both confirm the same inverting 3-bit DAC logic. The LTspice mountain shape is simply due to the specific timing of the pulse sources, which force a reset to zero. In contrast, the physical AD3 capture shows a continuous staircase because it uses a standard binary counter. The most important difference is saturation: the physical output hits a floor at -5V because the real Op-Amp

cannot output a voltage lower than its negative power rail. The simulation appears cleaner because it uses an ideal model that ignores this physical clipping. Despite these visual differences in timing and range, both results demonstrate the correct downward-stepping transfer function.

Graph of the Current showing it meets the constraint



#### b. Description of how the circuit works

The circuit in question is a 3-bit Binary-Weighted Resistor Digital-to-Analog Converter (DAC) designed to convert a 3-digit binary code, ranging from 000 to 111, into a specific voltage level between 0V and -5V. To ensure accurate signal processing, the circuit employs input buffers (U2, U3, U4), which have an extremely high input resistance of 500 M $\Omega$ . This feature allows them to read the logic voltage without drawing significant current from the digital source, only about 10 nA, thereby preventing any violation of current constraints.

The circuit uses three resistors ( $R1 = 10\text{k}\Omega$ ,  $R2 = 20\text{k}\Omega$ ,  $R3 = 40\text{k}\Omega$ ) to assign specific weights to each bit. The most significant bit (MSB) has the smallest resistance ( $10\text{k}\Omega$ ), allowing the most current to flow into the summing node, while the middle bit ( $20\text{k}\Omega$ ) allows half the current of the MSB, and the least significant bit (LSB) at  $40\text{k}\Omega$  permits only a quarter of the current. All three currents converge at the

inverting input (negative terminal) of the operational amplifier (Op-Amp), which maintains a virtual ground at 0V due to the non-inverting terminal being grounded.

In this configuration, the Op-Amp forces incoming currents to flow through a feedback resistor ( $R_F = 10\text{k}\Omega$ ), and according to Ohm's Law ( $V = I \times R$ ), the total current flowing through  $R_F$  results in the output voltage. The circuit is set up for inversion, meaning that a positive 5V input yields a negative output. For instance, when the MSB is 5V, the resulting output voltage is -5V, calculated by the equation  $V_{\text{out}} = -V_{\text{in}} \cdot (R_F/R_{\text{bit}})$ . Additionally, the Op-Amp requires dual power supplies, including a negative voltage (-Vee) from the Analog Discovery 3, to allow the output to swing negative. Without this dual supply setup, the Op-Amp would clip the output at 0V, preventing the accurate representation of the desired staircase voltage levels.

### c. Calculations

$$V_o = -R_F \left( \frac{V_1}{R} + \frac{V_2}{2R} + \frac{V_3}{4R} \right)$$

$$V_{\text{in}} = 5\text{V}$$

$$\text{Choose: } R = 5\text{k}\Omega$$

$$\text{MSB: } R = 10\text{k}\Omega$$

$$2R = 20\text{k}\Omega$$

$$\text{LSB: } 4R = 40\text{k}\Omega$$

$$\text{Step Size} = \frac{V_{FS}}{2^n} = \frac{5\text{V}}{2^8} = 0.625\text{V}$$

$$V_{\text{out}} = -V_{\text{in}} \cdot (R_F/R_{\text{bit}})$$

$$\text{MSB: } V = -5\text{V} \cdot (5\text{k}/10\text{k}) = -2.5\text{V}$$

$$\text{Middle bit} = -5\text{V} \cdot (5\text{k}/20\text{k}) = -1.25\text{V}$$

$$\text{LSB: } V = -5\text{V} \cdot (5\text{k}/40\text{k}) = -0.625\text{V}$$

Current Constraint:

$$I_{\text{input}} = V_{\text{logic}} / R_{\text{in}} = 5\text{V} / 500\text{M}\Omega = 10\text{nA}$$

Which is less than  $1\mu\text{A}$  the design constraint

This proves that the Resistor values are correct

| V1 | V2 | V3 | Digital Value | Vo (V) |
|----|----|----|---------------|--------|
| 0  | 0  | 0  | 0             | 0      |
| 0  | 0  | 1  | 1             | -0.625 |
| 0  | 1  | 0  | 2             | -1.250 |
| 0  | 1  | 1  | 3             | -1.875 |
| 1  | 0  | 0  | 4             | -2.5   |
| 1  | 0  | 1  | 5             | -3.125 |
| 1  | 1  | 0  | 6             | -3.750 |
| 1  | 1  | 1  | 7             | -4.375 |

Now, if we choose  $R_F = 10k\Omega$  with  $V_{in} = 3.3V$  for the physical setup, as we use Digital I/O pins

$$V_{out} = -V_{in} \cdot (R_F/R_{bit})$$

$$LSB: V = -3.3V \cdot (10k/40k) = -0.825V \text{ (STEP SIZE)}$$

| V1 | V2 | V3 | Digital Value | Vo (V)                    |
|----|----|----|---------------|---------------------------|
| 0  | 0  | 0  | 0             | 0                         |
| 0  | 0  | 1  | 1             | -0.825                    |
| 0  | 1  | 0  | 2             | -1.650                    |
| 0  | 1  | 1  | 3             | -2.475                    |
| 1  | 0  | 0  | 4             | -3.300                    |
| 1  | 0  | 1  | 5             | -4.125                    |
| 1  | 1  | 0  | 6             | -4.950                    |
| 1  | 1  | 1  | 7             | -5.775 (clipped to -5.00) |

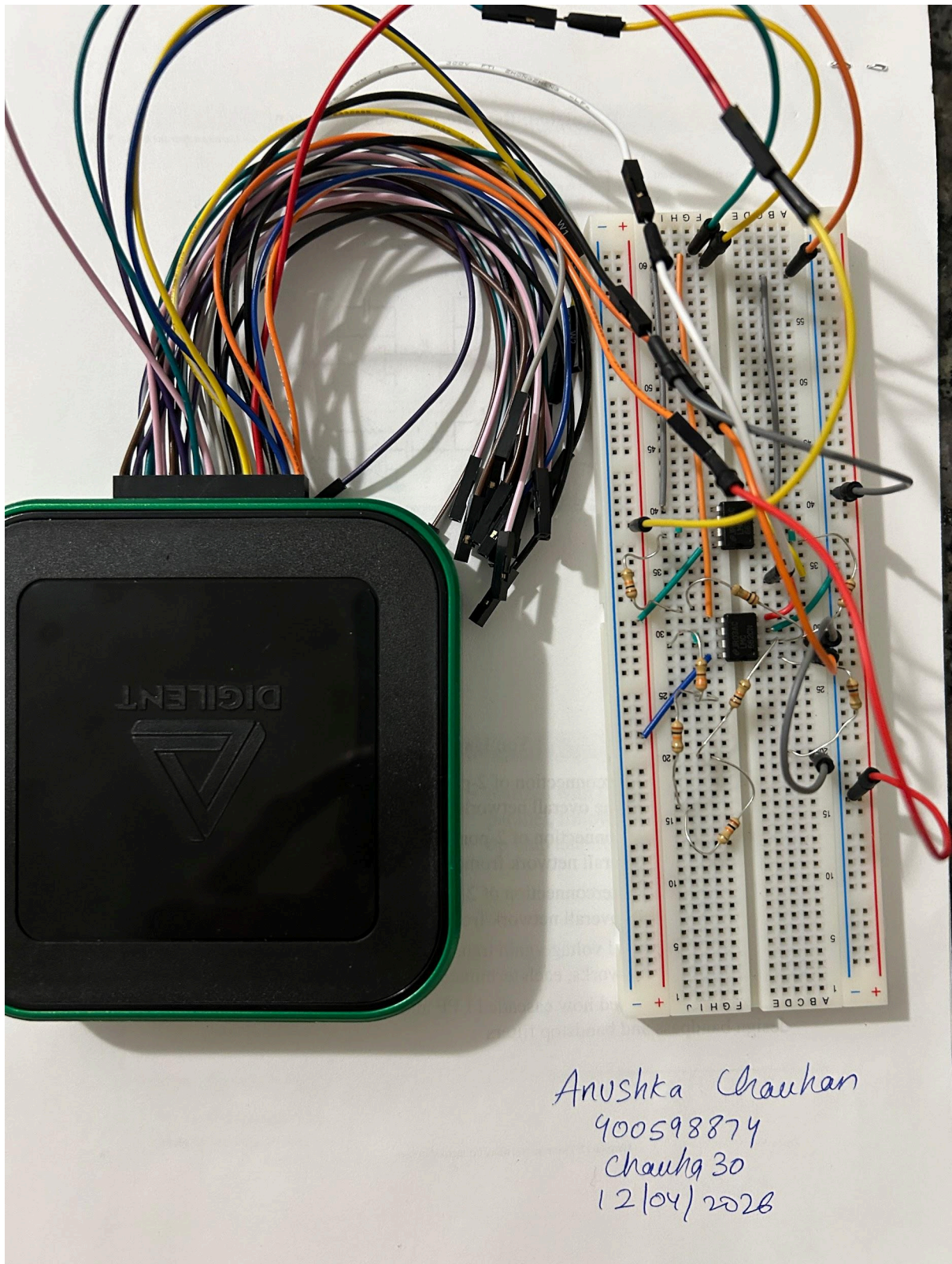
#### d. Reference for the Circuit

[1] N. K. Nikolova, "Lecture 2: Operational Amplifiers – Applications," course notes for CompEng 2CF3: Circuits and Waves for Computer Engineers, Dept. Elect. Comput. Eng., McMaster Univ., Hamilton, ON, Canada, 2026.

[2] FOKAL ACADEMY, "How to find Digital to Analog Converter Output Voltage || Opamp circuit Solved Problems || 3Bit DAC," YouTube, Oct. 8, 2023. [Video]. Available: <https://www.youtube.com/watch?v=JVkHEI5BB3w>. [Accessed: Apr. 10, 2026].

## Measurement and Analysis

### a. Physical Circuit



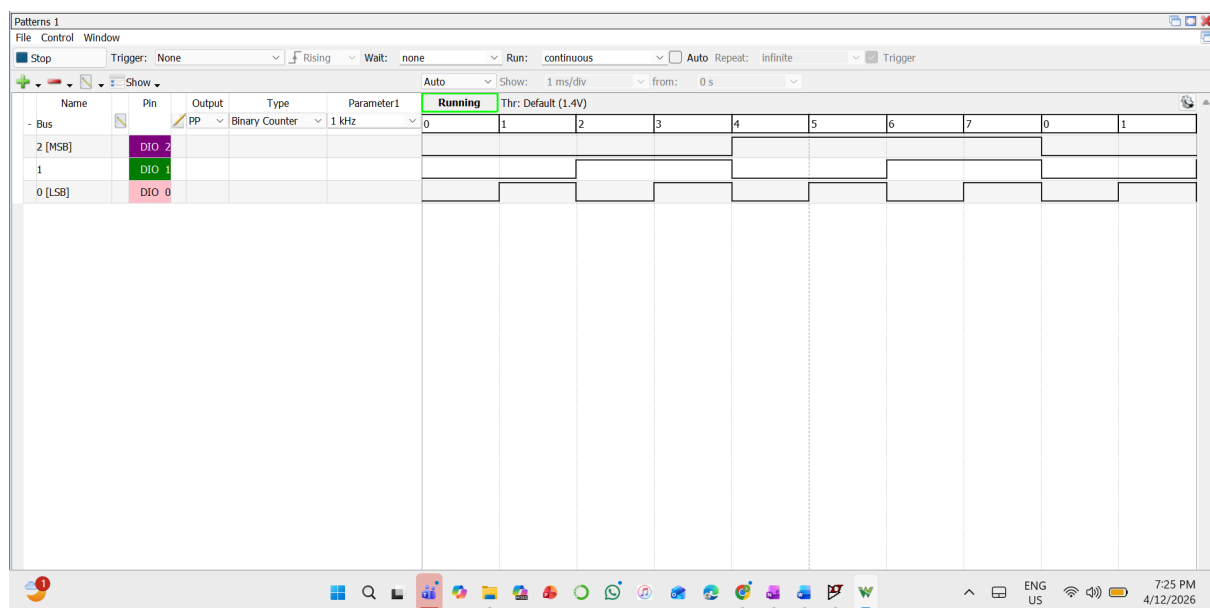
To ensure the physical circuit achieved the desired voltage range, a  $10\text{k}\Omega$  feedback resistor was used instead of the  $5\text{k}\Omega$  resistor specified in the initial simulation. This modification was

necessary because the Analog Discovery 3 (AD3) Digital I/O pins output a logic high of 3.3V, whereas the simulation utilized a 5V square wave. With the original 5k $\Omega$  resistor, the output voltage was limited to a maximum of only -3.3V. By increasing R<sub>F</sub> to 10k $\Omega$ , the circuit gain was doubled, allowing the analog output to effectively utilize the full -5V dynamic range provided by the Op-Amp's negative power supply rail (V<sub>ee</sub>).

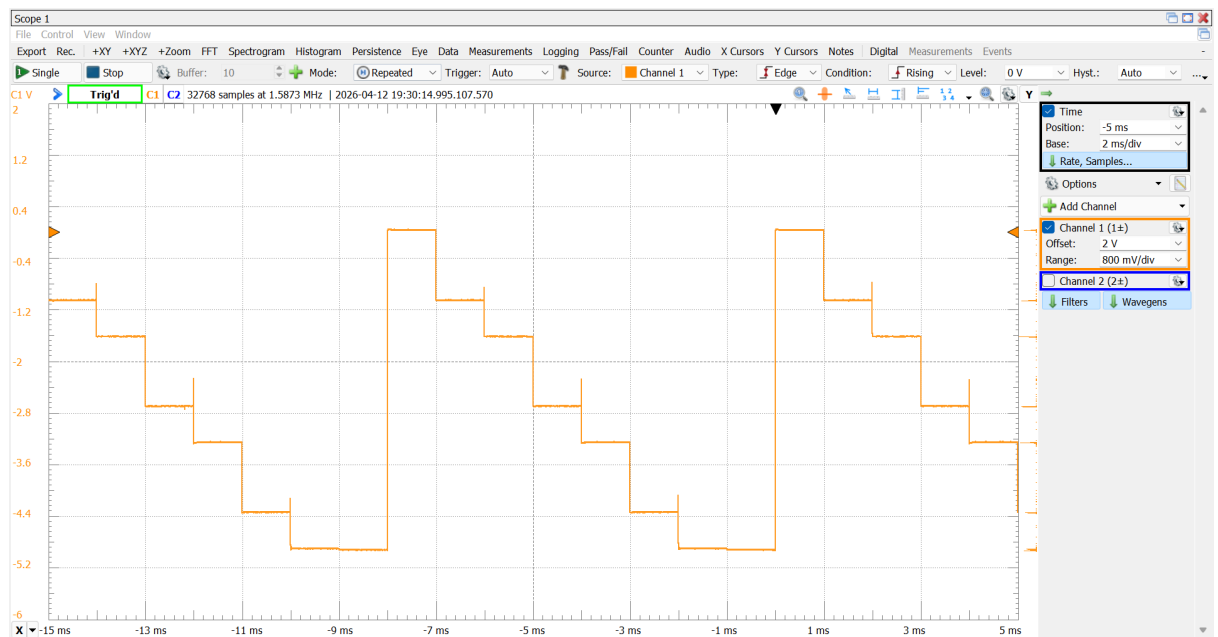
## b. Description of how the digital inputs were generated

To generate the 3-bit digital inputs for the D/A converter, the Pattern Generator tool within the Digilent WaveForms software was utilized. Digital I/O pins DIO 0, DIO 1, and DIO 2 on the Analog Discovery 3 were grouped into a 3-bit bus and configured as a Binary Counter.

This configuration automatically cycles through the eight binary states (000 to 111) in a perfectly synchronized sequence. The counter frequency was set to 1 kHz, allowing for a continuous, stable staircase waveform to be captured on the oscilloscope. The digital logic high voltage was configured to 5.0V to align with the design calculations for the weighted resistor network. By using the AD3's dedicated digital pins, the input signals provided clean, fast-switching logic transitions that were then isolated by the LMC662CN input buffers to ensure zero loading on the digital source.



### c. Transfer Characteristics of the Circuit



The transfer characteristic of the DAC describes how the 3-bit digital input is mapped to the analog output voltage. In this implementation, the circuit produces a monotonic, inverting staircase waveform. As the digital input increments from 000 to 111, the output voltage decreases in discrete, equal steps. Based on the 3.3V logic input and the 10kΩ feedback resistor, each binary count corresponds to a step size of approximately 0.825V. This step size is consistent across the first six transitions, showing excellent linearity and confirming that the weighted resistor network is correctly proportioning the current.

As the digital count reaches the final states, the output approaches the hardware limit of the -5V supply rail. While the theoretical calculation for the maximum digital input (111) is -5.775V, the physical Op-Amp saturates at the negative supply voltage. Consequently, the last step of the staircase appears slightly flattened or clipped at -5V. This saturation is an expected characteristic of the hardware setup, demonstrating that the DAC is utilizing the maximum possible output swing allowed by the Analog Discovery 3 power configuration.

### d. Gain Error

Gain error is the difference between the actual slope of the DAC's transfer function and the ideal theoretical slope. It indicates how accurately the circuit scales the digital input to the analog output. In this specific circuit, gain error is primarily caused by the 1% tolerance of the resistors and slight variations in the AD3's actual logic high voltage level.

The gain error was measured by isolating the Most Significant Bit (MSB) to avoid cumulative errors from other bit combinations. The digital input was set to 100 (MSB High, others Low). The resulting output voltage was measured using the Scope Measure tool to get the average to obtain a precise DC value. The measured average was -3.26554. This value was compared to the theoretical MSB target of -3.300V.

$$\text{Gain Error(\%)} = ((V_{\text{actual}} - (-3.30\text{V})) / -3.30\text{V}) \times 100\%$$

$$\text{Gain Error(\%)} = ((-3.26554 - (-3.30\text{V})) / -3.30\text{V}) \times 100\% = -1.044\%$$

The measured gain error is approximately -1.04%. This is an excellent result, as it falls within the expected range dictated by the 1% resistor tolerances.

#### e. Maximum Differential Non-Linearity

Differential Non-Linearity (DNL) measures the consistency and uniformity of the individual voltage steps in the DAC staircase. Ideally, every time the digital input increments by 1 LSB, the analog output should change by the same voltage amount (step size). DNL error occurs when these steps are unequal, usually due to mismatches in the 10k $\Omega$ , 20k $\Omega$  and 40k $\Omega$  resistor ratios. A DNL of 0 is perfect; if DNL is ever less than -1, the DAC is considered non-monotonic, meaning the voltage could actually move in the wrong direction. Based on the previously measured MSB of -3.26554V (which represents 4 steps), the ideal step size for this specific hardware is calculated as  $3.26554/4 = 0.81638$  V.

Now to find the actual Step DAC was set to 010(Binary 2). The output was measured using the Scope Measure tool to get the Average, yielding -1.61244V. Since this measurement is for 2 steps, the actual average step is  $1.61244/2 = 0.80622$ .

$$\text{DNL} = (\text{Vactual} - \text{Videal}) / \text{Videal} = (0.80622 - 0.81638) / 0.81638 = -0.0124 \text{ LSB}$$

The maximum DNL was found to be approximately -0.012 LSB. This extremely low value indicates that the steps are highly uniform. Since the error is well within the  $\pm 1$  LSB range, the DAC is perfectly monotonic, ensuring that every digital increment results in a clear and consistent analog decrease. This confirms that the 20k $\Omega$  and 40k $\Omega$  resistors are well-matched to the 10k $\Omega$  feedback resistor.

#### f. Offset Error

Offset error is the analog output voltage measured when the digital input is set to zero. Ideally, the output should be exactly 0V because no current is being summed at the inverting node of the Op-Amp. In a physical circuit, however, a small non-zero voltage often exists. This is typically caused by the input offset voltage of the Op-Amp (internal transistor mismatches), input bias currents flowing through the feedback resistor, or small grounding offsets between the AD3 and the breadboard. The measurement was conducted by setting all three Digital I/O lines (DIO 0, 1, and 2) to a Constant Low state in the Patterns tool. To capture the precise value of this error, the Oscilloscope vertical range was increased to a more sensitive setting. The Average measurement tool was utilized to average out high-frequency noise and thermal fluctuations, providing a stable DC offset reading.

Theoretical Vout at 000: 0V

Measured Vout at 000: 2.7582 mV

$$\text{Offset Error} = \text{Vmeasured} - \text{Vtheoretical} = 2.7582\text{mV}$$

Since this error is significantly smaller than the LSB step size(825mV), it has a negligible impact on the overall accuracy of the DAC and does not interfere with the monotonicity of the transfer characteristics.

## Discussion

### a. Origin of various errors measured

The discrepancies between the simulated ideal staircase and the physical measurements (Gain, DNL, and Offset) originate from real-world hardware non-idealities. The Gain Error of -1.04% is primarily a result of the Analog Discovery 3 (AD3) providing a 3.3V logic high rather than the ideal 5V assumed in standard models; while the feedback resistor was increased to 10k $\Omega$  to compensate, the gain remains sensitive to the precise voltage of the DIO pins and the 1% tolerance of the resistors. This resistor mismatch also causes the Differential Non-Linearity (DNL), as the binary-weighted current contributions do not sum to perfectly equal step sizes. Furthermore, while simulation shows a mathematically perfect staircase, the physical capture exhibits output saturation at -5V, proving the Op-Amp cannot drive beyond its negative power rail. Finally, the Offset Error of 2.76 mV stems from the LMC662CN's internal input offset voltage, which is omitted in ideal simulations but appears in physical hardware even when all inputs are logic low.

### b. Design Choice and Alternate Design

The Binary-Weighted Resistor design was selected for this 3-bit implementation due to its conceptual simplicity and minimal component count, making it ideal for demonstrating fundamental current-summing principles. To satisfy the design constraint of drawing less than 1 $\mu$ A from the digital bit lines, Op-Amp buffers were integrated at the input stage. As verified by the LTspice simulation and the LMC662 datasheet, the extremely high CMOS input impedance of these buffers reduces the current draw to the pico-amp range, effectively isolating the AD3 from the DAC's resistor network. An alternative design considered was the R-2R Ladder. The primary advantage of the R-2R architecture is that it requires only two resistor values (R and 2R), regardless of the DAC's resolution. This eliminates the need for the wide range of precision binary-weighted resistors required by the current design, which becomes physically impractical and prone to massive linearity errors in higher-bit systems.

## References

- [1] Tutorials Point, "Linear Integrated Circuits - Digital to Analog Converters," <https://www.google.com/search?q=tutorialspoint.com>. [Online]. Available: [https://www.tutorialspoint.com/linear\\_integrated\\_circuits\\_applications/linear\\_integrated\\_circuits\\_applications\\_digital\\_to\\_analog\\_converters.htm](https://www.tutorialspoint.com/linear_integrated_circuits_applications/linear_integrated_circuits_applications_digital_to_analog_converters.htm). [Accessed: Apr. 12, 2026].
- [2] Electronics Course, "Digital to Analog Converter (DAC)," *electronics-course.com*. [Online]. Available: <https://electronics-course.com/digital-analog-converter>. [Accessed: Apr. 12, 2026].
- [3] Analog Devices, "How to Adjust and Calibrate Offset and Gain Error in Precision Digital-to-Analog Converter (DAC) Systems," *analog.com*, Technical Article. [Online].

Available:

<https://www.analog.com/en/resources/technical-articles/adjust-and-calibrate-offsetgain-error-in-precision-dac-calculation--maxim-integrated.html>. [Accessed: Apr. 12, 2026].

[4] Microchip Technology, "Offset and Gain Error Correction," *microchip.com*, Support Article. [Online]. Available: <https://support.microchip.com/s/article/Offset-and-Gain-Error-Correction>. [Accessed: Apr. 12, 2026].

[5] Wikipedia contributors, "Differential nonlinearity," *Wikipedia, The Free Encyclopedia*. [Online]. Available: [https://en.wikipedia.org/wiki/Differential\\_nonlinearity](https://en.wikipedia.org/wiki/Differential_nonlinearity). [Accessed: Apr. 12, 2026].

[6] "Analog Discovery 3 - Digilent Reference," *digilent.com*. Accessed April 12, 2026. [Online]. Available: <https://digilent.com/reference/test-and-measurement/analog-discovery-2/start?redirect=1>

[7] N. K. Nikolova, "Lecture 2: Operational Amplifiers – Applications," course notes for CompEng 2CF3: Circuits and Waves for Computer Engineers, Dept. Elect. Comput. Eng., McMaster Univ., Hamilton, ON, Canada, 2026.

[8] FOKAL ACADEMY, "How to find Digital to Analog Converter Output Voltage || Opamp circuit Solved Problems || 3Bit DAC," YouTube, Oct. 8, 2023. [Video]. Available: <https://www.youtube.com/watch?v=JVkHEI5BB3w>. [Accessed: Apr. 10, 2026].

[9] Texas Instruments, "LMC662 CMOS Dual Operational Amplifier datasheet," *ti.com*, Rev. March 2013. [Online]. Available: <https://www.ti.com/product/LMC662>. [Accessed: Apr. 12, 2026].